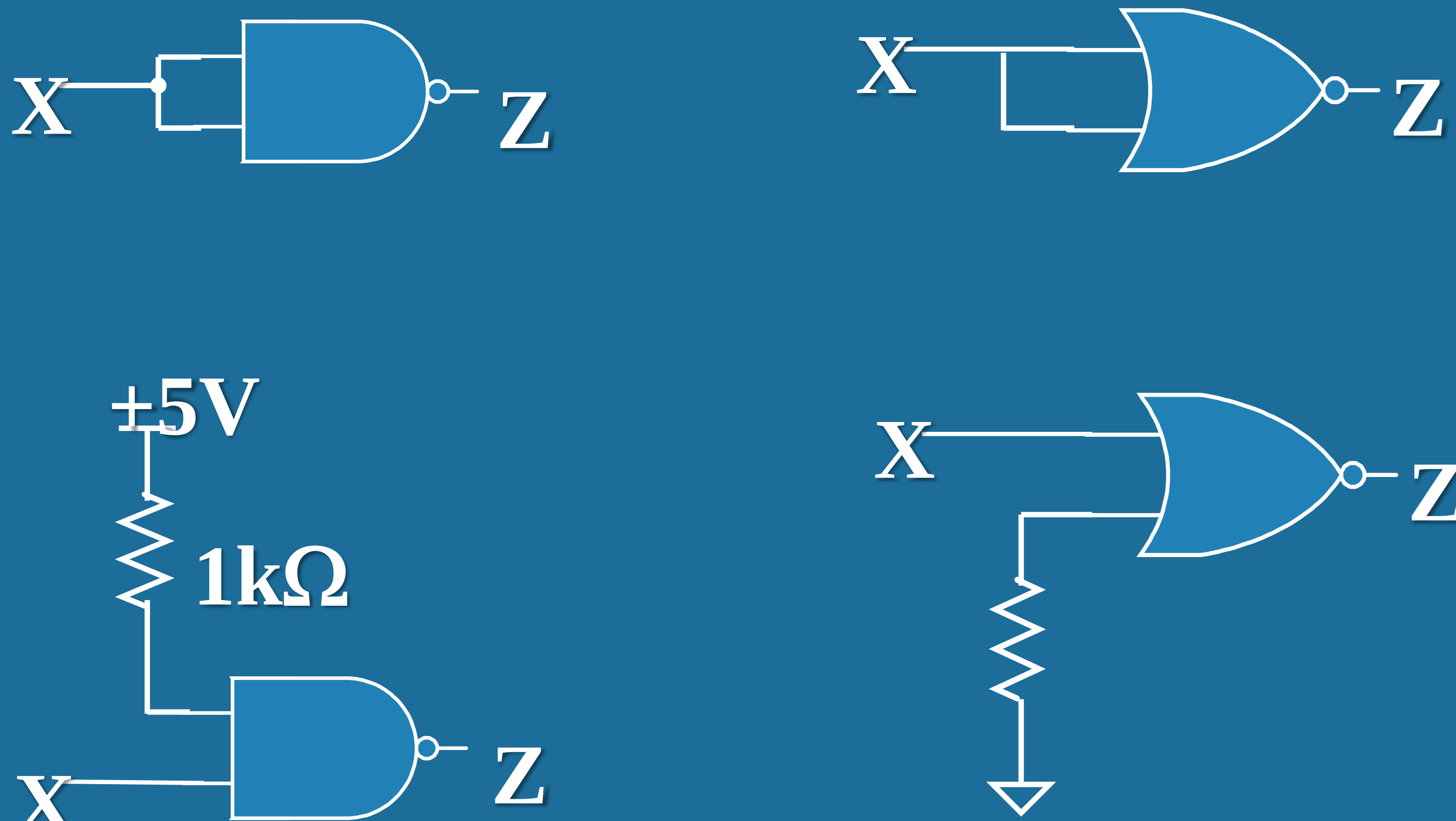
第三章作业 答案

T1： Draw the diagrams of three ways to deal with unused inputs of CMOS circuit.

T1: Unused inputs of CMOS circuit



T2: What are the three states of the three-state buffer?

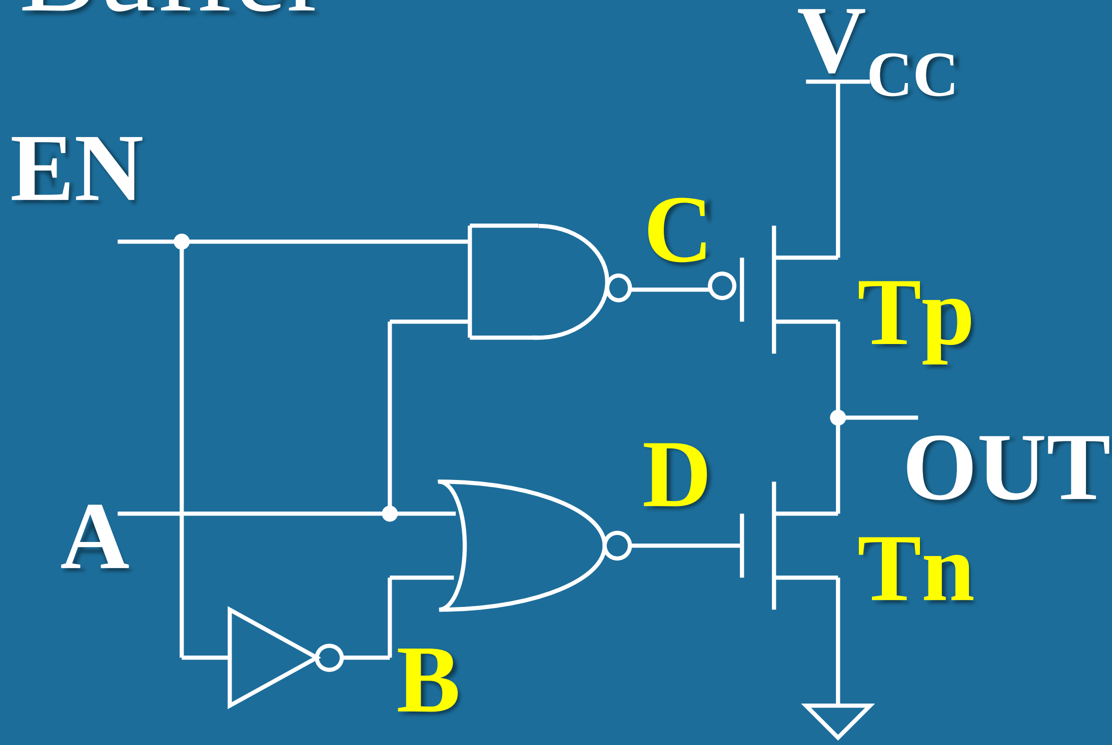
T2: Three-state Buffer

EN=0,

C=1, Tp is OFF

B=1, D=0, Tn is OFF

OUT is floating state



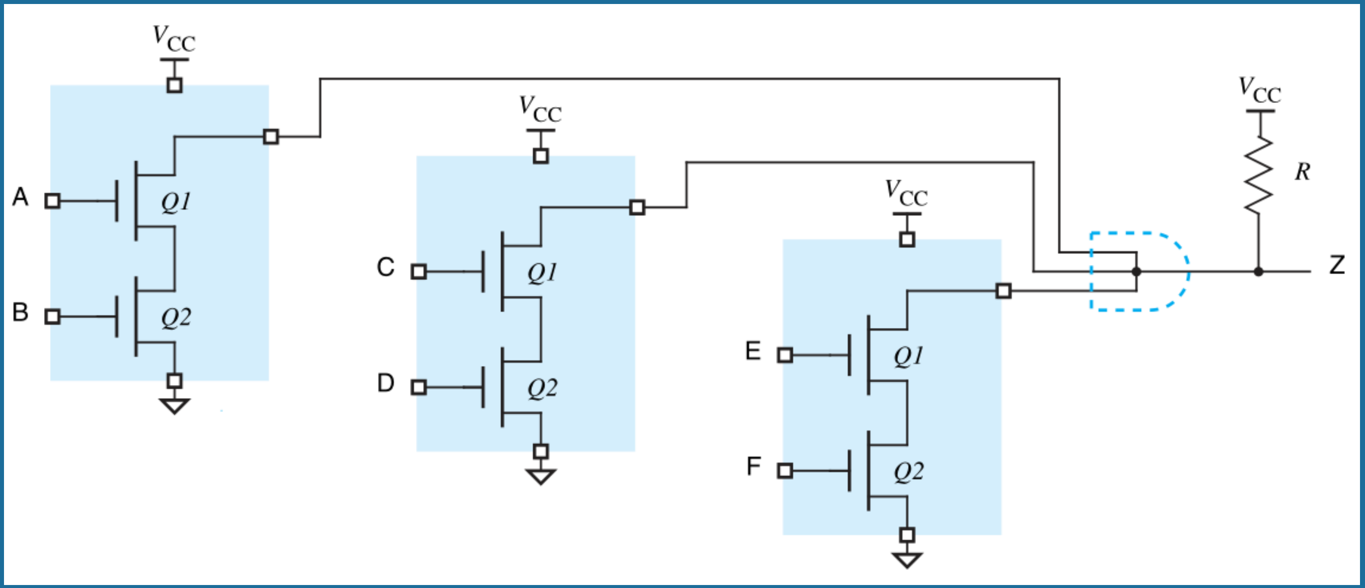
EN=1,

C=$\overline{A}$, B=0 , D=$\overline{A}$

OUT=A

OUT is low-state or high-state

T3: Write the logic function of Z in the following figure. What is the name for the circuit connection in the blue dotted line area of the figure?

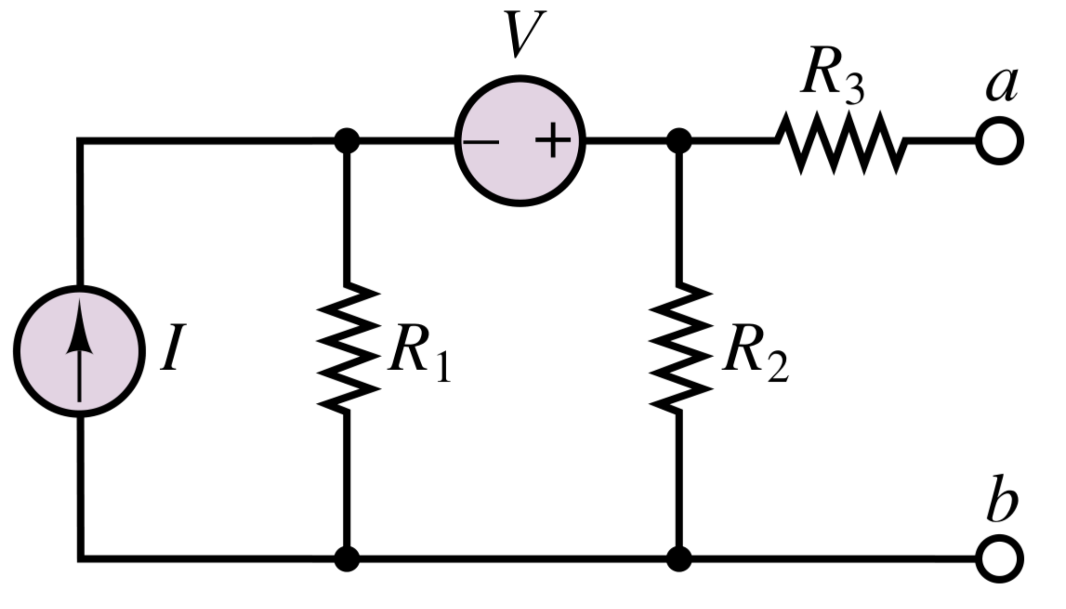


$Z = \overline{AB + CD + EF}$

Name: "Wired AND"

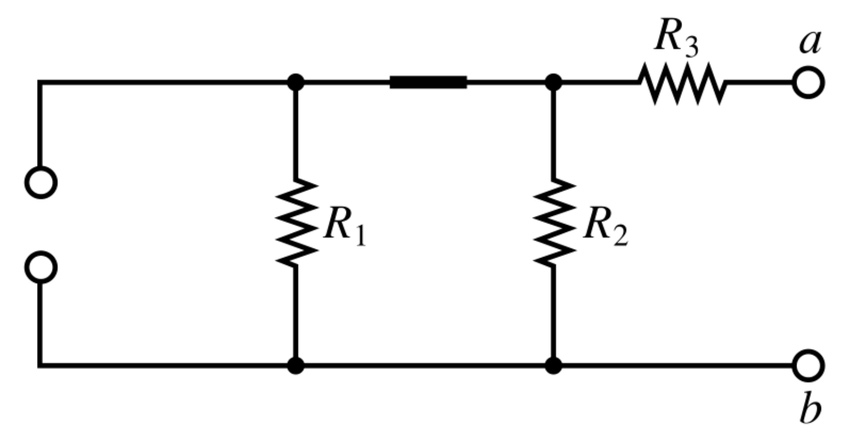
T4： Determine the Thevenin and Norton equivalents for left side of port a-b of the circuit below.

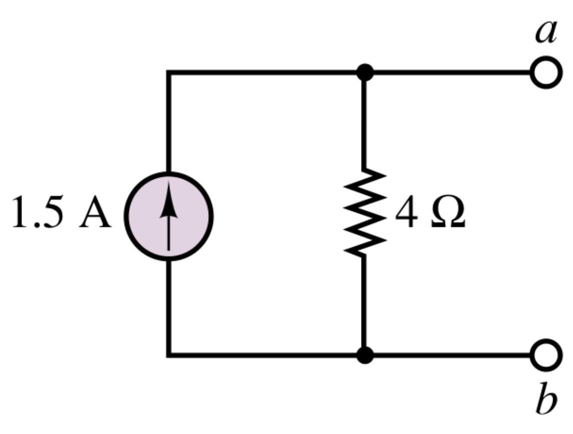
$V = 6$ V; $I = 2$ A; $R_1 = 6\Omega$; $R_2 = 3\ \Omega$; $R_3 = 2\ \Omega$



$$V_{ab} = V_I + V_v = 2A \times \frac{3\Omega \times 6\Omega}{3\Omega + 6\Omega} + V \times \frac{3\Omega}{3\Omega + 6\Omega} = 4V + 2V = 6V$$

$$R_T = R_1 \| R_2 + R_3 = 6\|3 + 2 = 4\ \Omega$$





$$R_T = R_1 \| R_2 + R_3 = 6 \| 3 + 2 = 4\ \Omega$$

$$V_{ab} = V_I + V_v = 2A \times \frac{3\Omega \times 6\Omega}{3\Omega + 6\Omega} + V \times \frac{3\Omega}{3\Omega + 6\Omega} = 4V + 2V = 6V$$

$$R_T = R_N = R_{eq}$$

$$I_N = \frac{Vab}{R_T} = \frac{6V}{4\Omega} = 1.5A$$